

```
import numpy as np
```

```
#1.random module
```

```
#(1a).randint()-->random+integer
#it will create an array in which it will provide random numbers in agiven range.
```

```
#syntax=>np.random.randint(min_num,max_num,total)
```

```
a=np.random.randint(1,10,3)
a
```

```
array([5, 1, 1])
```

```
#(1b) seed()=>it will fix your random genreted data
```

```
np.random.seed(45)
a=np.random.randint(1,10,3)
a
```

```
array([4, 1, 6])
```

```
#(1c)rand()=>it will genreted num in range 0-1
```

```
a=np.random.rand(5)
a
```

```
array([0.4444695 , 0.47280797, 0.048522 , 0.16332445, 0.11595071])
```

```
#2 linspace(min,max,total_num)==>
#it will create an aaray in which it will provide same step num value in a given range
```

```
a=np.linspace(1,2,5)
a
```

```
array([1. , 1.25, 1.5 , 1.75, 2.  ])
```

```
#11-50 mid 4 random num genreted
#array-->[7,7,7,7,7,7,7]
#2-d [11-19] array --->shape,size,dimension
```

```
a=np.random.randint(11,50,4)
a
```

```
array([19, 34, 19, 23])
```

```
a=np.full(8,7)
a
```

```
array([7, 7, 7, 7, 7, 7, 7, 7])
```

```
a=np.array([[11,12,13],
            [14,15,16],
            [17,18,19]])
a
```

```
array([[11, 12, 13],
       [14, 15, 16],
       [17, 18, 19]])
```

```
# reshaping the array
# n(rows)*n(col)=n(total_element)
```

```
a=np.array([1,2,3,4,5,6,7,8,9,10,11,12])
a
```

```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12])
```

```
#1*12 ,12*1,2*6,6*2,3*4,4*3
```

```
a.reshape(2,6)
```

```
array([[ 1,  2,  3,  4,  5,  6],  
       [ 7,  8,  9, 10, 11, 12]])
```

```
a.reshape(3,4)
```

```
array([[ 1,  2,  3,  4],  
       [ 5,  6,  7,  8],  
       [ 9, 10, 11, 12]])
```

```
a.reshape(6,5)
```

```
-----  
ValueError                                Traceback (most recent call last)  
/tmp/ipykernel_985/2497865584.py in <cell line: 0>()  
----> 1 a.reshape(6,5)
```

```
ValueError: cannot reshape array of size 12 into shape (6,5)
```

Next steps: [Explain error](#)

Start coding or [generate](#) with AI.